Between the Lines: a computational analysis of migration narrative in *The New York Times*

WS 25/26

Computational Social Science

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Research questions

How does a major media institution like *The New York Times* approach a sensitive topic like migration?

How has this approach changed and evolved over the last 20 years (2005–2025)?

Has the *The New York Times* narrative been influenced by the political background and presidential administrations?

Outline

- Data
- Exploratory lexical analysis
- Sentiment analysis
- Topic modelling
- Multiple linear regression model and inference
- Conclusions

1. Data

DATASET EXTRACTION

- **NYT Archive API**
- 2005-2025
- Selected variables

“Date” , “Headline”, “Abstract”, “Author”, “News section”.

- Filtered by

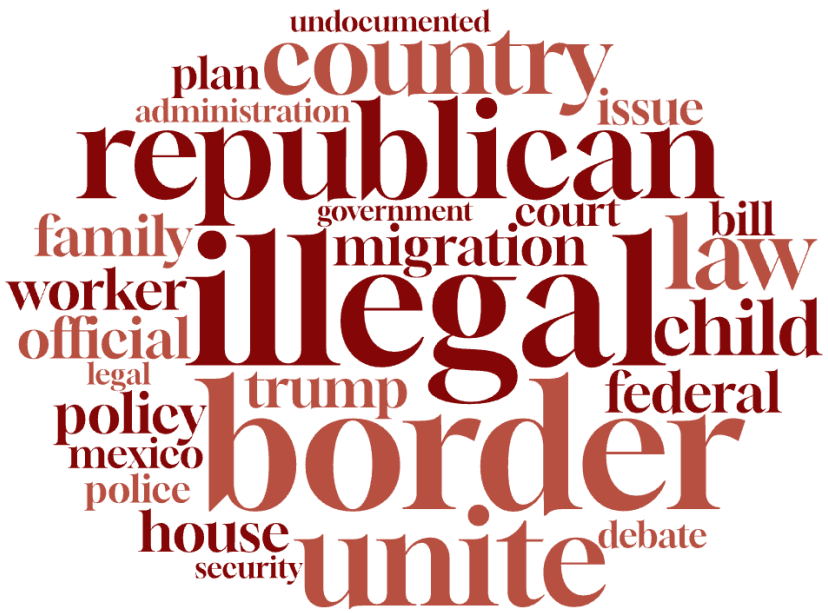
“Migration”, “Migrations”, “Immigrant”, “Immigrants”, “Migrate”, “Immigrate”.

DATASET PREPROCESSING

- Outliers removal: deletion of abstracts with words number >200.
- Noise Removal: deletion of archive metadata, page numbers and duplicated abstracts.
- Formatting text for consistent Sentiment Analysis and Topic Modelling .

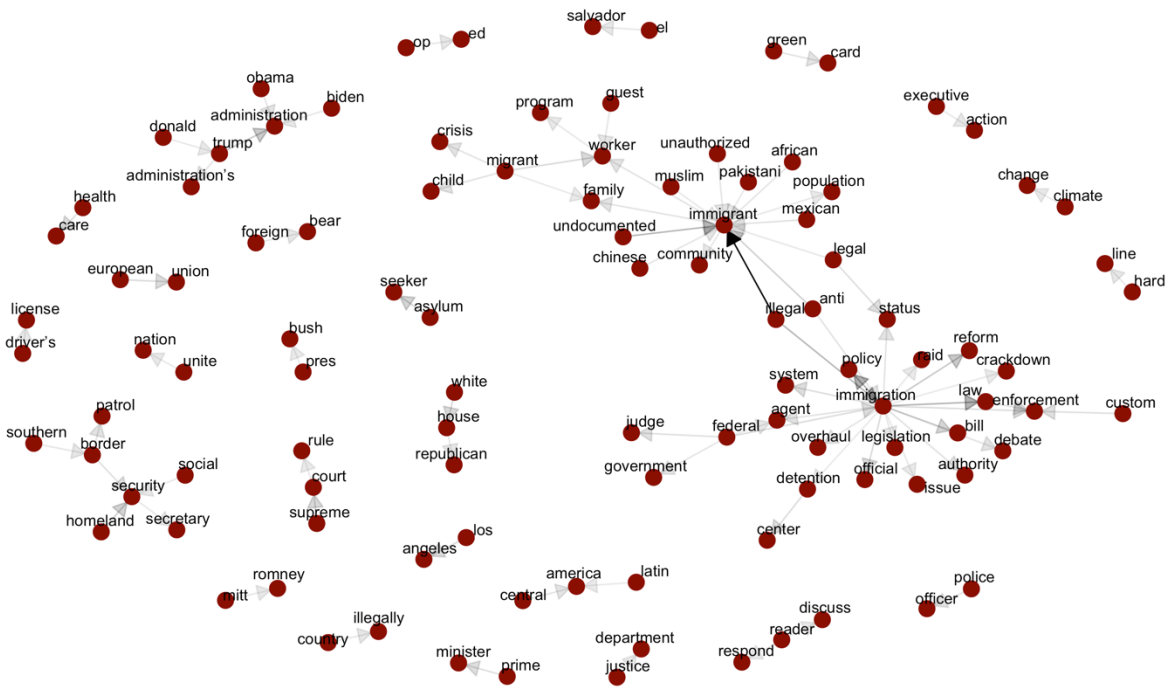
2. Exploratory lexical analysis

Word cloud



1. illegal
2. border
3. unite
4. country
5. law
6. trump
7. federal
8. policy
9. worker
10. official

Bigrams network graph



1. illegal immigrant
2. immigration policy
3. immigration law
4. undocumented immigrant
5. illegal immigration
6. immigration reform
7. trump administration
8. anti immigrant
9. homeland security
10. asylum seeker

3. Sentiment Analysis

Dual approach:
Bing vs. AFINN

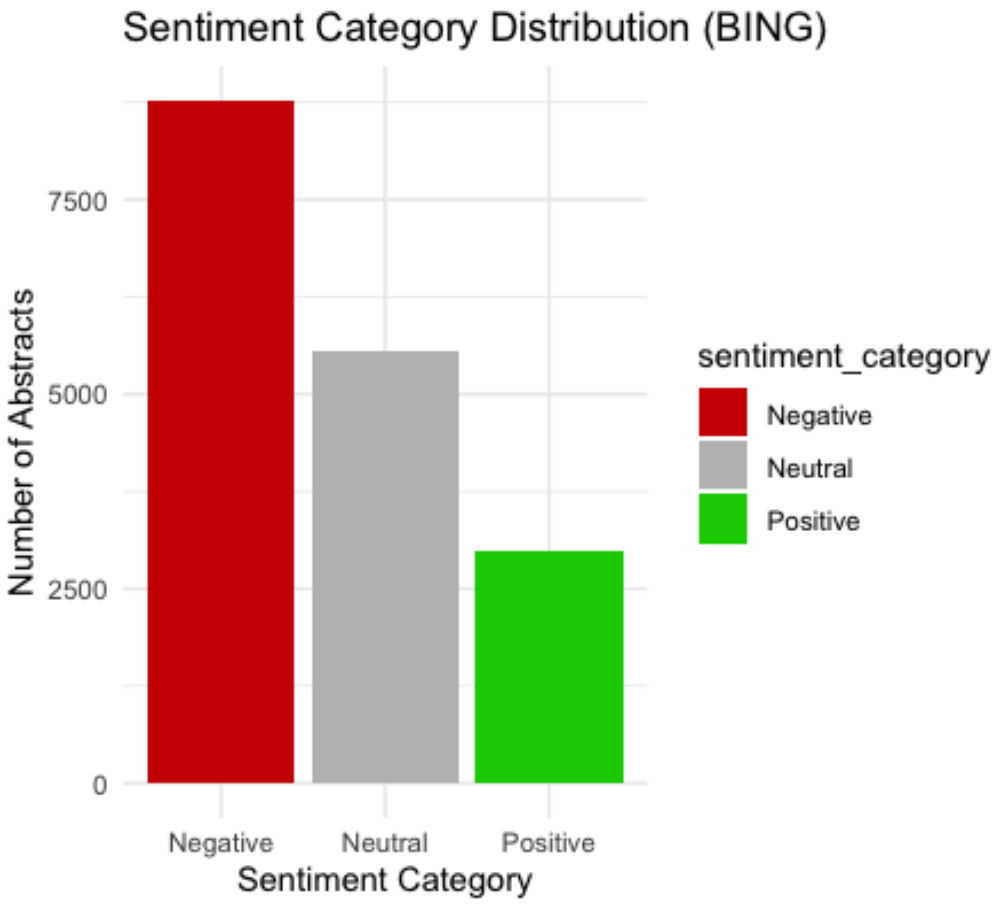
Bing Lexicon: A binary classification (Positive/Negative) used to identify the volume of polarized abstracts.

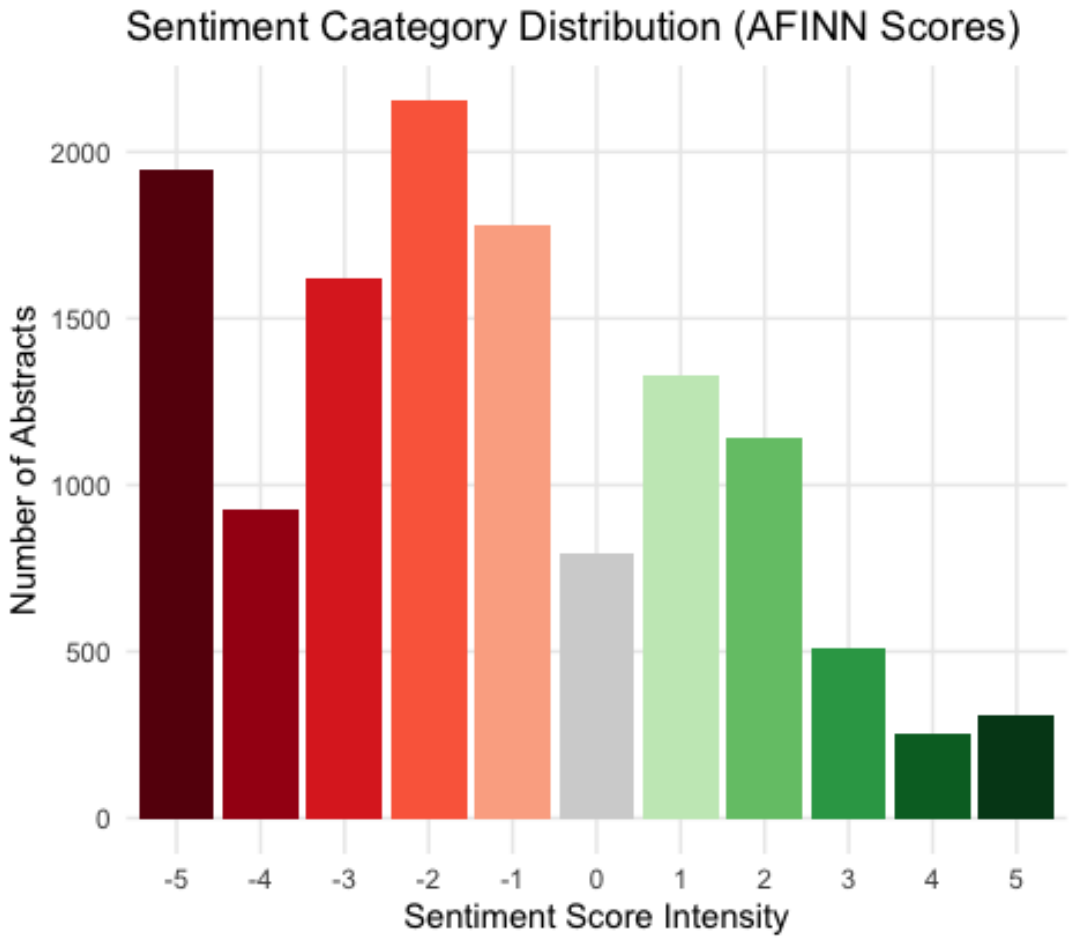
AFINN Lexicon: An intensity-based approach (scores from -5 to +5) used to measure the "weight" of the emotional tone.

Why?

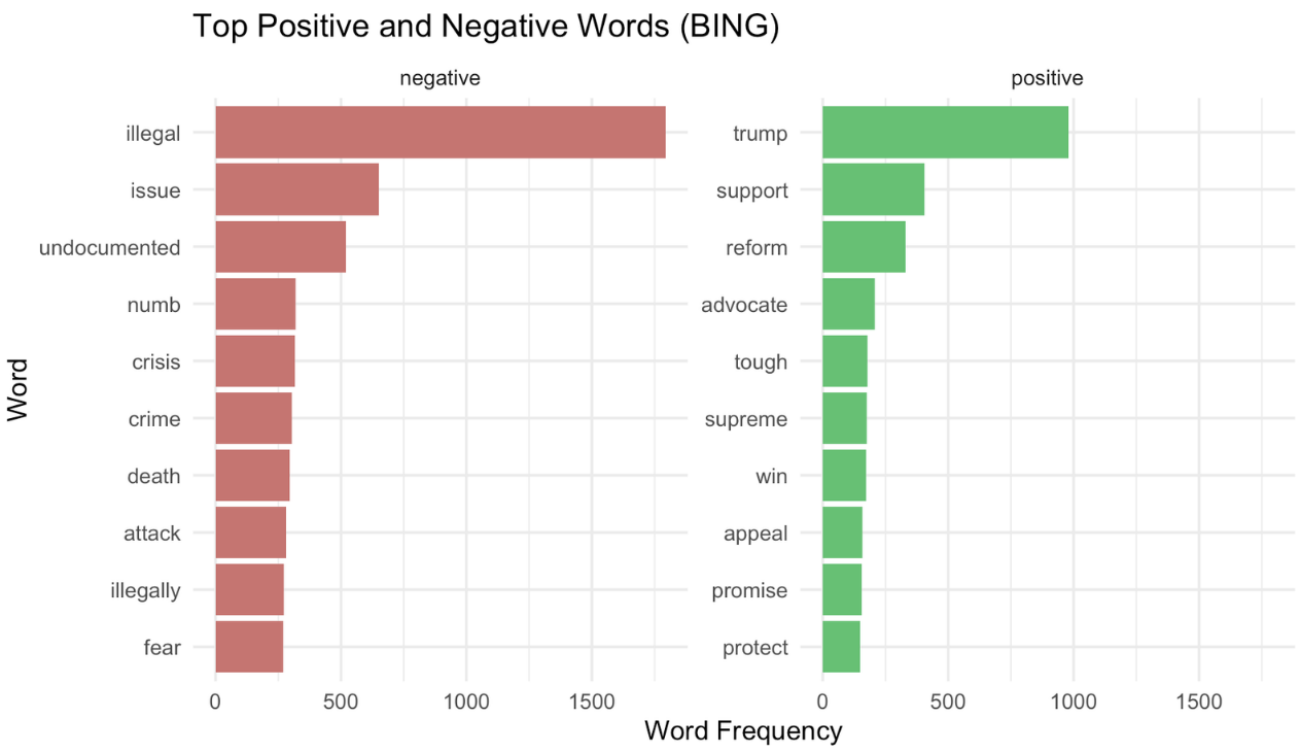
- Using two different dictionaries allows for the cross-validation of results; if both lexicons show the same trend, the findings are significantly more reliable and robust.
- Capturing different intensities with AFINN.

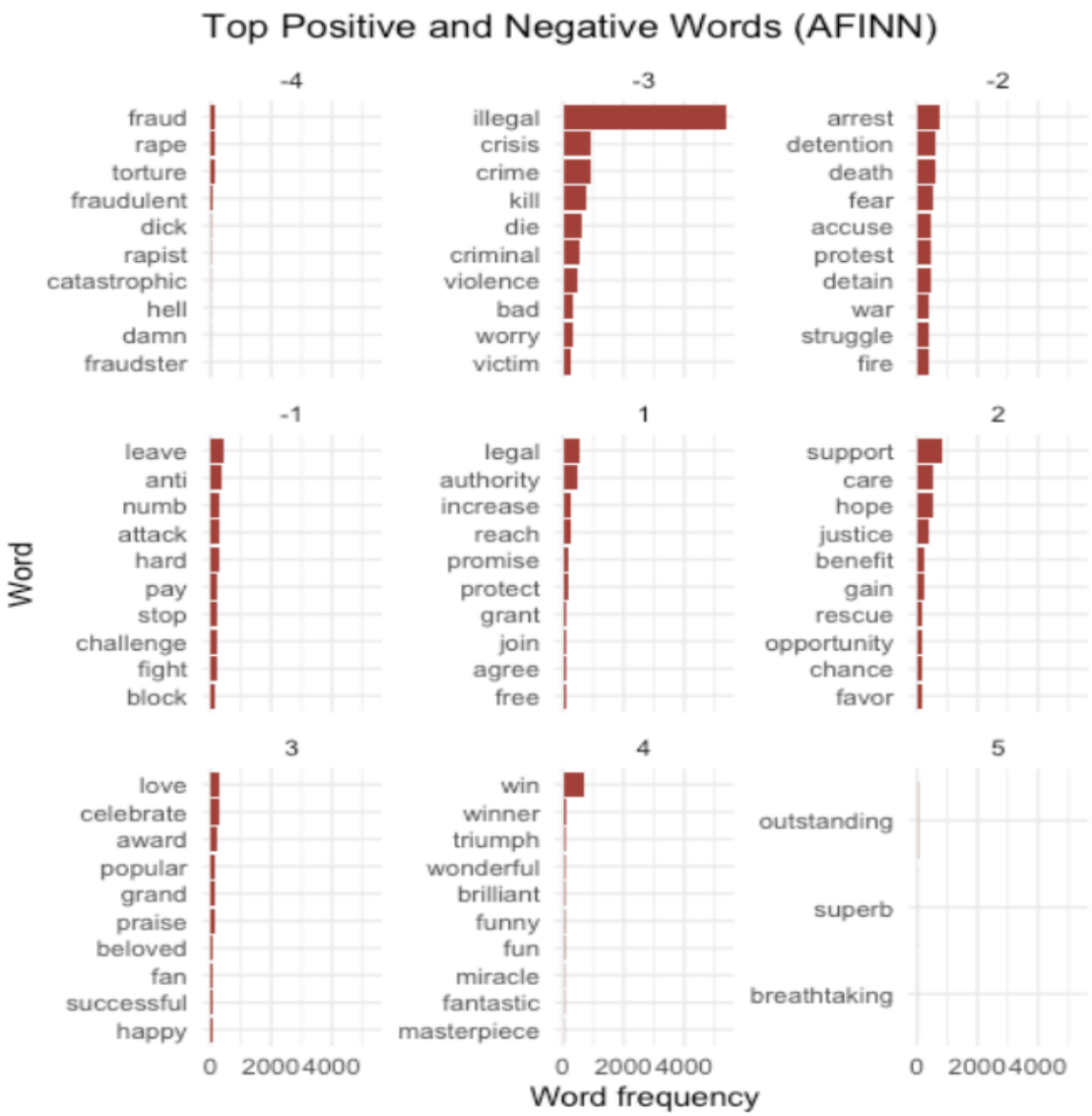
3. Sentiment Analysis



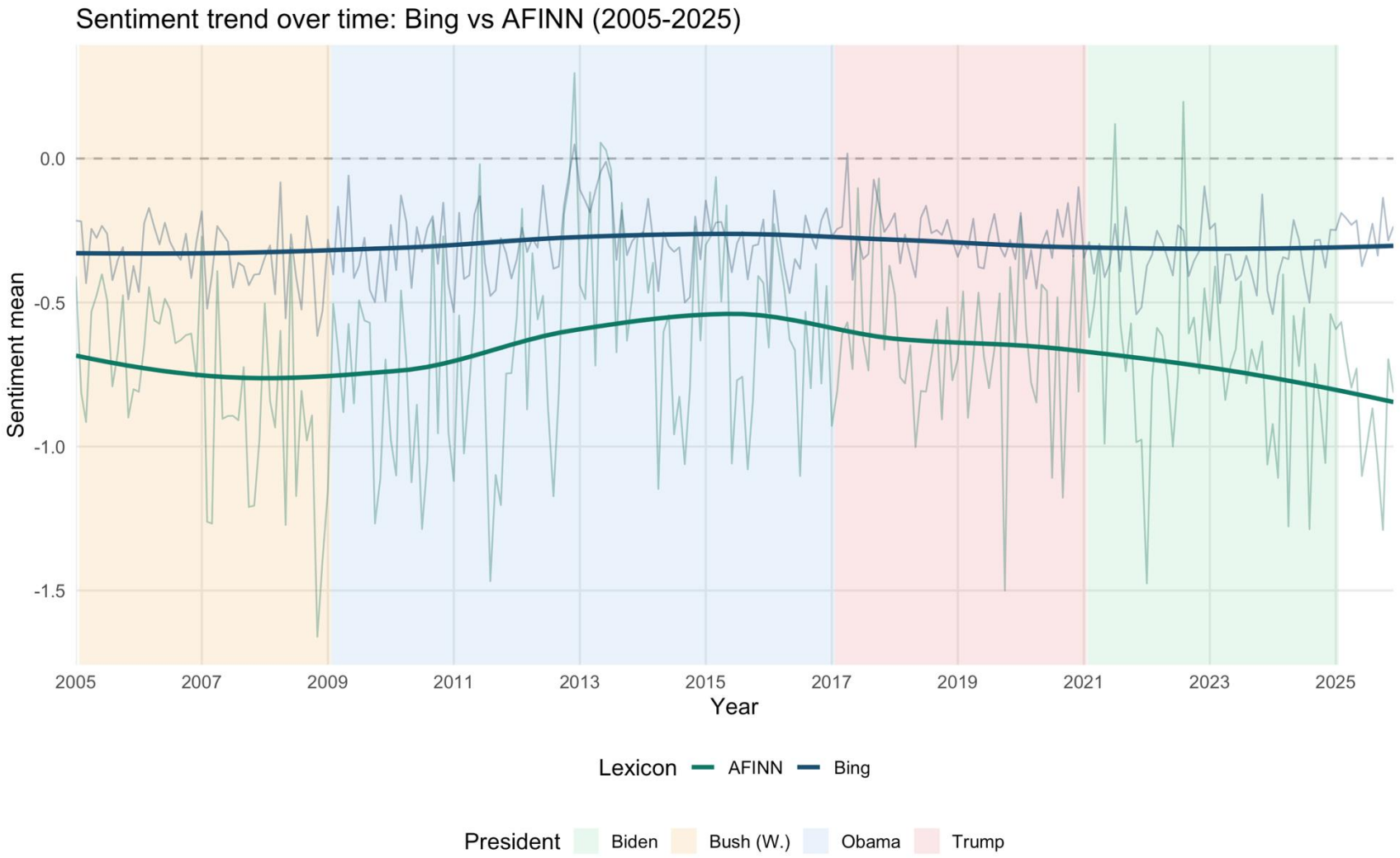


3. Sentiment Analysis





3. Sentiment Analysis - Trend over time



- Both Bing and AFINN lexicons show a **remarkably stable, predominantly negative trend**.
- **AFINN captures more variance and intensity**, but both lines follow the same trajectory.
- The sentiment mean does **not** show **significant shifts when transitioning between** Republican (Bush, Trump) and Democratic (Obama, Biden) **administrations**.

3. Sentiment Analysis

Multiple Linear Regression Model

$$\text{sentiment_mean}_i = \beta_0 + \beta_1 \text{year}_i + \beta_2 \text{president}_i + \beta_3 \text{word_count}_i + \varepsilon_i$$

Results:

Predictor	Estimate (β)	Std. Error	t value	Pr(>|t|)	Significance
(Intercept)	1.6713	7.9155	0.211	0.833	
Year	-0.0009	0.0039	-0.246	0.806	
President (Obama)	-0.0176	0.0305	-0.578	0.563	
President (Trump)	0.0256	0.0511	0.502	0.616	
President (Biden)	-0.0772	0.0685	-1.126	0.260	
President (Trump II)	0.0103	0.0784	0.132	0.895	
Word Count	**-0.0307**	**0.0033**	**-9.129**	**< 2e-16**	*******

F-statistic: $p < 2.2*10^{(-16)}$ - **Adjusted R^2**: 0.008

- **Word count significance:** the negative coefficient indicates that longer abstracts are associated with more negative sentiment.
- **Presidential stability:** no statistically significant shift in the *New York Times* tone compared to the Bush baseline.
- **Global significance:** Although the R^2 is low, the **F-statistic** confirms the model is overall significant, proving that these findings are not due to random chance.

3. Sentiment Analysis

Test ANOVA

$$H_0 : \mu_{Bush} = \mu_{Obama} = \mu_{Trump} = \mu_{Biden} \qquad H_0 : \beta_{word_count} = 0 \qquad H_0 : \beta_{year} = 0$$

$$SS_{Total} = SS_{Model} + SS_{Residuals} \qquad F = \frac{MS_{Model}}{MS_{Residuals}} = \frac{SS_{Model}/df_{Model}}{SS_{Residuals}/df_{Residuals}}$$

Results:

Source of Variation	Df	Sum Sq (SS)	Mean Sq (MS)	F value	$Pr(>F)$	Decision
Year	1	0.1	0.078	0.1646	0.685	**Fail to Reject H_0**
President	4	12.3	3.066	6.4394	**3.581e-05**	**Reject H_0 *****
Word Count	1	39.7	39.677	83.3419	**< 2.2e-16**	**Reject H_0 *****
Residuals	12753	6071.3	0.476			

- **Presidential era** significally influences the overall variance of the reporting.
- The **Year** variable is not significant, fluctuations based on specific events.
- President variable: **model vs ANOVA**

4. Topic Modelling

- **Text pre-processing**: custom stop-word removal (e.g., "migration", "NYT"), tokenization and creation of DTM (Document-Feature Matrix).

- Applied **LDA (Topic Modeling)** on abstract text to detect 5 hidden thematic clusters.

TOPIC 1
Border and crisis flows

```
"border"
"countri"
"mexico"
"tri"
"thousand"
"asylum"
"govern"
"cross"
"near"
"arriv"
```

TOPIC 2
Political and legislative debate

```
"trump"
"polici"
"republican"
"hous"
"issu"
"bill"
"nation"
"plan"
"illeg"
"senat"
```

TOPIC 3
Daily life and integration

```
"famili"
"children"
"live"
"way"
"center"
"work"
"school"
"life"
"student"
"street"
```

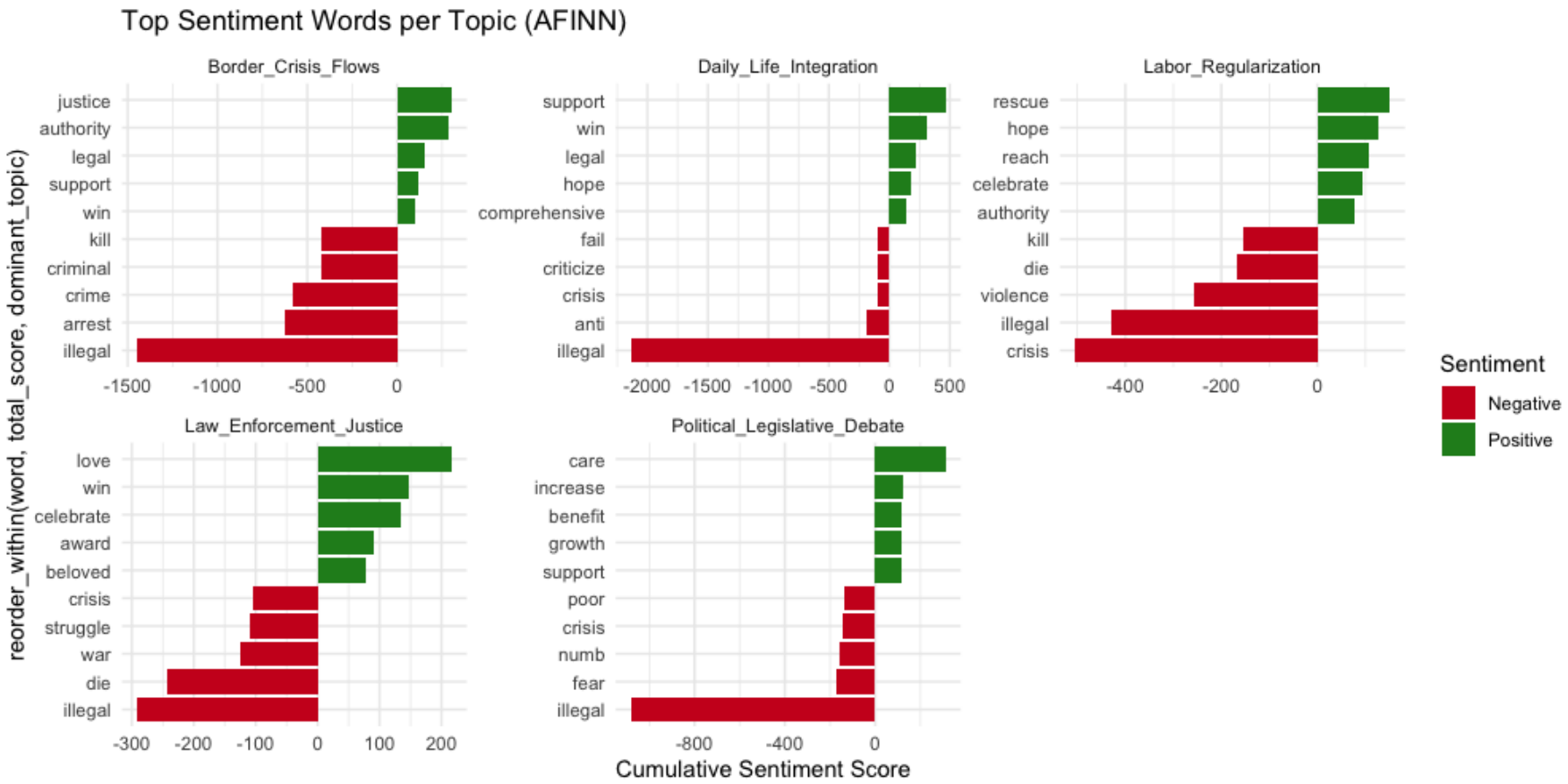
TOPIC 4
Law enforcement and justice

```
"feder"
"law"
"offici"
"deport"
"state"
"court"
"enforc"
"polic"
"right"
"author"
```

TOPIC 5
Labour and regularization

```
"illeg"
"worker"
"group"
"legal"
"work"
"home"
"number"
"program"
"chang"
"recent"
```

4. Topic Modelling x Sentiment Analysis



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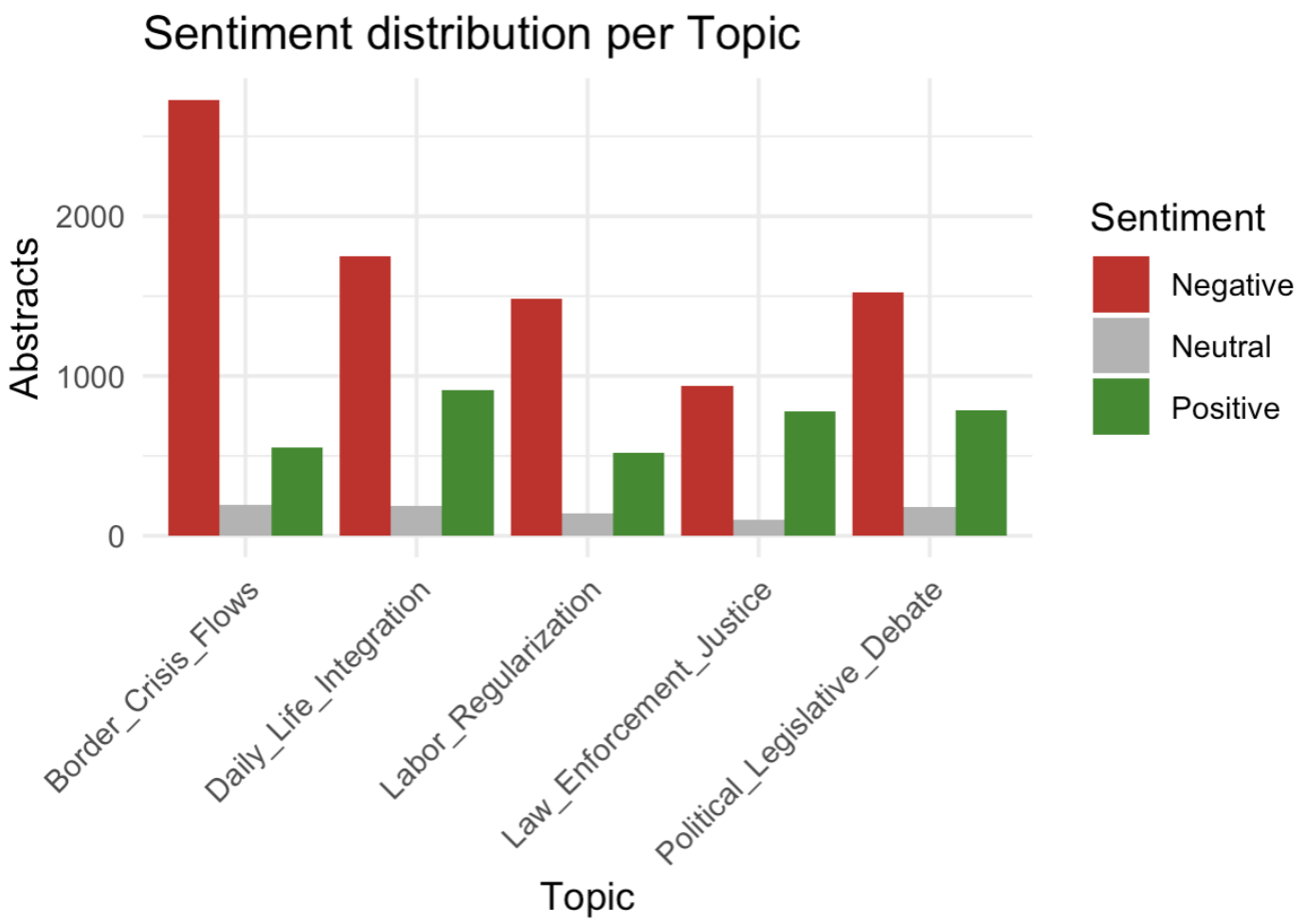
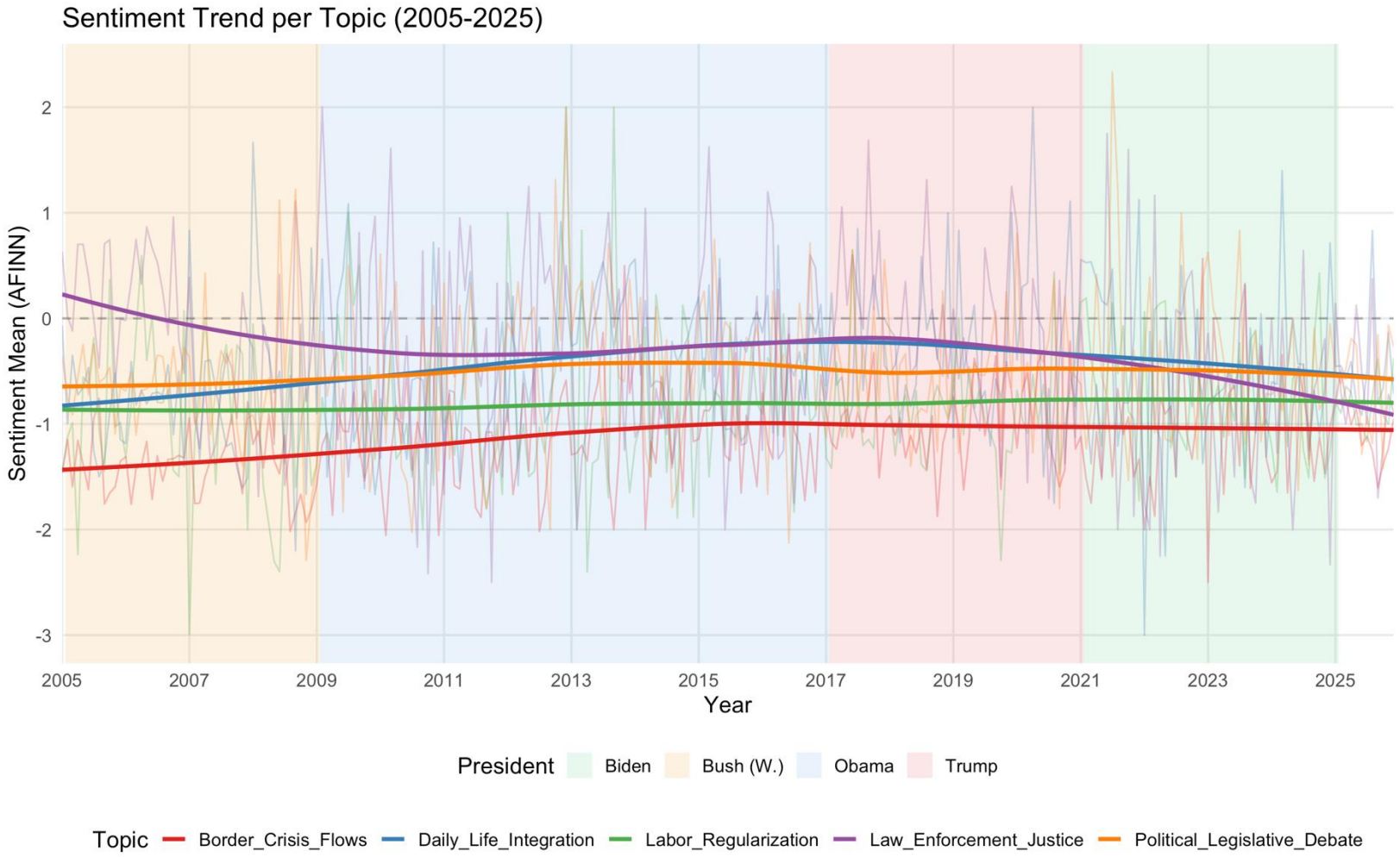


4. Topic Modelling x Sentiment Analysis – Trend over time



- **Border and Crisis Flows** is consistently the most negative topic across all years
- **Law Enforcement and Justice** shows the most significant downward trajectory, from the most neutral theme to a high-intensity negative frame.
- **Most topics** (Legal, Labor, Political) **have converged** toward a similar negative mean.
- **Presidential transitions** seems to not cause big trend reversals.

4. Topic Modelling

Multiple Linear Regression Model

$$Sentiment = \beta_0 + \beta_1(Year) + \beta_2(President) + \beta_3(WordCount) + \beta_4(Topic) + \epsilon$$

Results:

F-statistic: $p < 2.2*10^{(-16)}$ - **Adjusted R^2**: 0.0398

Predictor	Estimate (β)	Std. Error	t value	Pr(>|t|)	Significance
(Intercept)	-4.2366	7.8430	-0.540	0.589	
Year	0.0018	0.0039	0.486	0.627	
President (Obama)	-0.0050	0.0301	-0.169	0.866	
President (Trump)	0.0399	0.0505	0.791	0.429	
President (Biden)	-0.0738	0.0676	-1.091	0.275	
Word Count	**-0.0285**	**0.0033**	**-8.588**	**< 2e-16**	***
Topic: Daily Life	**0.2066**	**0.0174**	**11.872**	**< 2e-16**	***
Topic: Law Enforc.	**0.3460**	**0.0198**	**17.391**	**< 2e-16**	***
Topic: Political	**0.1309**	**0.0179**	**7.293**	**3.22e-13**	***

- **Higher R^2**, adding topics to the model significantly increased its explanatory power: specific subject matter as primary driver of the article's tone.
- Positive coefficients for **Daily Life, Law Enforcement and Political Debate** indicate that these themes are **statistically "less negative"** than the baseline topic, Border and Crisis Flows (border-related news most critical part of the narrative).
- **Word Count** remains a highly significant negative predictor.
- **Presidential administrations** (Obama, Trump, Biden) remains statistically non-significant

4. Topic Modelling

Test ANOVA

$$H_0 : \mu_{Bush} = \mu_{Obama} = \mu_{Trump} = \mu_{Biden}$$

$$H_0 : \mu_{Topic1} = \mu_{Topic2} = \mu_{Topic3} = \mu_{Topic4} = \mu_{Topic5}$$

$$H_0 : \beta_{word_count} = 0$$

$$H_0 : \beta_{year} = 0$$

$$SS_{Total} = SS_{Model} + SS_{Residuals}$$

$$F = \frac{MS_{Model}}{MS_{Residuals}} = \frac{SS_{Model}/df_{Model}}{SS_{Residuals}/df_{Residuals}}$$

Results:

Source of Variation	Df	Sum Sq (SS)	Mean Sq (MS)	F value	$Pr(>F)$	Decision (H_0)
Year	1	0.1	0.078	0.1700	0.6801	**Fail to Reject**
President	4	12.3	3.066	6.6526	**2.41e-05**	**Reject *****
Word Count	1	39.7	39.677	86.1014	**< 2.2e-16**	**Reject *****
Dominant Topic	4	196.4	49.108	106.5674	**< 2.2e-16**	**Reject *****
Residuals	12749	5874.9	0.461			

- The test identifies the **dominant Topic** as the single most influential factor in the model, followed by **article length**.
- It reveals that the **Presidential era** introduces a statistically significant level of variance, even if no single president represents a radical change from the long-term editorial baseline.

5. Conclusions

Findings

- **Structural negativity:** The *New York Times* migration narrative is characterized by a persistent negative sentiment. This is not a temporary trend but a structural feature of how the topic is framed.

- **Topic-driven sentiment:** the analysis found that the specific theme, such as border security or legal enforcement, is the most powerful predictor of sentiment.

- **The "Depth" paradox:** increased investigation does not lead to a more balanced view; instead, it consistently correlates with more critical and severe reporting.

- **Political stability:** despite the polarized American landscape, the *NYT* editorial line remains remarkably stable across the Bush, Obama, Trump, and Biden eras, showing no significant partisan shifts in its core sentiment.

5. Conclusions

Limitations and Future Research

- This computational framework is **highly reproducible** and can be used to analyze thousands of articles across multiple decades and different national contexts.

- The current study focuses exclusively on *The New York Times*: **expanding the analysis to include a diverse editors** would allow for a more accurate comparative study of partisan media framing.

- Abstracts provide a concentrated version of the narrative**, analyzing full-text articles would offer a better understanding of the linguistic complexity and emotional dept**h of the coverage.

- **Comparing newspaper narratives with social media discourse** could reveal how immigration sentiment differs between institutional journalism and public perception.

Thanks for the attention!